Detecting Heart Failure

Early-in-Stay Identification from Temporal EHR Laboratories with a Testosterone Case Study

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***Abstract*—Early-in-stay identification of admissions with heart failure (HF) can support timely clinical decisions, reduce downstream complications, and improve allocation of limited hospital resources. Electronic health records (EHR) enable large-scale modeling for HF, but real-world deployment is challenged by missing and irregular measurements, label noise, and selection effects in test ordering. In this work, we predict whether an admission is coded with HF using ICD code prefixes (ICD-10 I50* and ICD-9 428*), using only laboratory data from the first 48 hours of the admission. We study testosterone as a biologically plausible biomarker whose inpatient measurement is rare and clinically selective, and we evaluate whether it provides incremental signal beyond routine laboratory panels. We begin with a minimal, interpretable baseline under strict memory constraints, then extend to a temporal laboratory representation that better reflects early admission physiology. We emphasize leakage-safe evaluation, threshold tuning for imbalanced outcomes, quantitative interpretability, and robustness analyses (ablations, calibration, occlusion) to ensure conclusions are methodologically defensible. Across these analyses, we find that temporal modeling of routine laboratory panels provides substantial gains over a testosterone-only baseline, while testosterone numeric values offer limited marginal benefit at population level due to extreme sparsity and non-random ordering.**

***Index Terms*—heart failure, MIMIC-IV, electronic health records, laboratory time series, testosterone, LSTM, imbalanced classification, interpretability, calibration**

I. Introduction

Heart failure (HF) is a major cause of morbidity, mortality, and hospital utilization. Even when HF is not the primary admission diagnosis, early physiologic changes and comorbidity patterns can indicate elevated risk. Predictive models built from EHR can support earlier recognition, triage, and targeted follow-up, especially when deployed as decision-support tools rather than automated decision-makers.

A. Clinical Task Definition: Early-in-Stay Identification

A key point is the clinical task we address. We do not model future HF onset in a population without known disease. Instead, we predict whether an admission is coded with HF (ICD-10 I50* and ICD-9 428*), using only laboratory measurements available in the first 48 hours of that admission. This setting is best described as early-in-stay identification (or phenotyping under an early window), rather than long-horizon prediction of incident HF.

Because ICD codes can reflect a pre-existing condition or history documented during the stay, onset time is not modeled. Our labels capture coded HF association with the admission, not necessarily acute HF onset within the first 48 hours. We explicitly treat this as a limitation and avoid causal claims.

B. Clinical and Data Background

From a clinical viewpoint, HF is heterogeneous: it can present as acute decompensation, chronic progression, or secondary complications driven by renal dysfunction, anemia, infection, or arrhythmia. This heterogeneity is mirrored in EHR signals: different subtypes and care pathways produce different measurement patterns. For example, some admissions include intensive metabolic panels, troponin measurements, and frequent complete blood counts, while others include only sparse labs. As a result, models can inadvertently learn patterns of care (which tests are ordered) rather than underlying physiology, unless missingness is explicitly represented and carefully analyzed.

From a data viewpoint, EHR observations are irregularly sampled and missing not at random. Laboratory tests are ordered because clinicians suspect specific conditions, and ordering behavior is influenced by specialty, availability, and hospital protocols. This introduces selection effects that can distort both model training and interpretation. Therefore, robust early-in-stay HF identification requires: (i) time-aware feature restrictions to avoid leakage, (ii) evaluation beyond accuracy, and (iii) interpretability analyses that can separate value-driven signal from ordering-driven signal.

C. Motivation: Testosterone as a Case Study Biomarker

Testosterone has been studied in relation to cardiovascular outcomes, including HF severity and prognosis, but evidence is nuanced and confounded [16], [17]. From a modeling viewpoint, testosterone is a useful case study because it is biologically plausible but rarely measured in general inpatient workflows. This creates a practical gap between biomarker plausibility and EHR utility. If testosterone is ordered only for a small, clinically selective subset, (i) the numeric value may carry limited population-level signal, and (ii) the measurement indicator itself can encode selection bias. A robust evaluation should therefore test both numeric values and measurement patterns, and should avoid over-interpreting performance differences without ablations.

D. Study Overview and Contributions

We structure the work as a progression from a minimal baseline to robust temporal modeling.

Stage 1: Interpretable baseline (testosterone-focused). We materialize a testosterone-measured cohort from MIMIC-IV v3.1 [1] using a memory-bounded streaming strategy. We train interpretable baselines (logistic regression and GPU XGBoost) on three clinically motivated covariates (testosterone, age, sex). This stage exposes practical pitfalls in EHR ML (threshold sensitivity under imbalance and the risk of biased sampling during early data ingestion shortcuts).

Stage 2: Temporal lab modeling (early window). We expand the feature set to a panel of 89 blood-like analytes and encode irregular labs in the first 48 hours into a fixed-length sequence (12 buckets of 4 hours). We train an LSTM sequence model [5] and compare it with strong tabular baselines.

Robust evaluation and scientific conclusion. We enforce patient-level splits, tune thresholds on validation, report AUPRC in addition to AUROC, and provide quantitative interpretability and robustness (SHAP, ablations, calibration, group occlusion). The key scientific conclusion is not that testosterone is clinically irrelevant, but that in MIMIC-IV its measurement is too sparse and selective to improve population-level early-in-stay HF identification beyond routine labs.

II. Related Work

We summarize prior work along four axes: (i) HF modeling using structured EHR, (ii) temporal modeling and missingness, (iii) interpretability and probability quality, and (iv) testosterone and cardiovascular outcomes.

A. HF Modeling from Structured EHR

Structured EHR models for clinical endpoints often use demographics, diagnoses, medications, vitals, and labs as predictors. Large-scale efforts demonstrate that deep learning can leverage heterogeneous EHR signals, though performance and interpretability depend on feature design and label definitions [10]. For HF, studies target incident HF, decompensation, readmission, or mortality, and use labels derived from codes or clinical definitions [11]. While neural models can learn flexible representations, gradient boosting frequently remains a competitive baseline due to strong performance on tabular data and favorable compute cost [2].

Benchmarking work emphasizes that model comparisons are sensitive to preprocessing, missingness handling, and label timing [12]. In particular, it is easy to leak information when features include tests ordered after clinicians already suspect HF. This motivates our strict early-window restriction and patient-level splitting.

B. Temporal Modeling and Missingness

Clinical time series are irregular and sparse. A common approach is time discretization: represent each time bin using summary statistics such as last value, mean, and count. This creates fixed-size representations for classical models and can work well when the time horizon is short.

Neural approaches model sequences directly. Missingness-aware recurrent models incorporate observation masks and time gaps as explicit inputs and often outperform naive imputations [4]. Other methods emphasize interpretability, such as RETAIN, which applies reverse-time attention to generate explanations [13]. Transformer-based models for EHR event sequences (for example BEHRT) emphasize representation learning over long histories and heterogeneous event types [14]. However, these models require careful data alignment, robust missingness handling, and substantial compute; their advantages are most evident when data density and event variety are high.

C. Interpretability and Probability Quality

Interpretability is essential for clinical credibility. For tree ensembles, SHAP provides a principled additive attribution approach that supports both global and local explanations [3]. For deep models, ablations and occlusion tests provide model-agnostic evidence by measuring performance changes when specific signals are removed. These approaches can also reveal dataset biases, including test-ordering effects.

Probability quality is important when predicted risks are used for triage. Brier score and reliability analysis are standard tools for evaluating calibration [9]. In imbalanced settings, calibration and thresholding interact strongly, and decision thresholds should be selected based on the desired operating point and validation performance [7], [8]. For HF identification, AUPRC and operating-point metrics (precision/recall trade-offs) are particularly informative because they emphasize the positive class [7].

D. Testosterone and Cardiovascular Outcomes

Clinical literature suggests associations between testosterone, hypogonadism, and cardiovascular conditions, including HF, but conclusions vary across cohorts and adjustment strategies [16], [17]. From an EHR ML perspective, a key methodological point is that hormone assays are ordered under specific clinical suspicions. Therefore, measurement presence can correlate with specialty workflows or comorbidity patterns, and models can exploit this correlation. Without ablations, a model could incorrectly attribute predictive power to numeric hormone values when the real driver is the clinical pathway that led to testing. Our work explicitly evaluates both numeric values and measurement indicators to mitigate this risk.

A growing survey literature summarizes deep learning for EHR and highlights recurring pitfalls: label leakage, missingness that is not at random, and lack of external validation [15]. These concerns are especially relevant for biomarker-centric studies, where measurement itself can be a proxy for clinician intent. Our evaluation design (patient-level splits, early-window restriction, and ablations on measurement presence) is directly motivated by these prior observations.

Pipeline summary:
(1) Data access (MIMIC-IV v3.1)
(2) Cohort labeling (ICD codes)
(3) Materialization with streaming chunks
(4) Stage 1: testosterone-focused baseline features
(5) Stage 2: 48h temporal tensorization (values + masks)
(6) Models: LR, XGBoost/GB, LSTM
(7) Evaluation: patient-level splits, AUROC/AUPRC, threshold tuning
(8) Analysis: SHAP (trees), ablations, calibration, occlusion

Fig. 1. End-to-end pipeline used in this study.

III. Methodology

This section details our data processing, feature construction, models, and evaluation protocol. Figure 1 summarizes the end-to-end pipeline.

A. Data Source and Tables

We use MIMIC-IV v3.1 [1]. We rely on `labevents` for lab measurements, `d_labitems` for lab metadata, `patients` for demographics, and `diagnoses_icd` for ICD codes. All processing is performed on de-identified data with access obtained through credentialing.

B. Label Definition and Temporal Restrictions

Stage 1 (patient-level baseline). A patient is labeled HF-positive if any ICD-10 code begins with "I50". This coarse label is used for a minimal testosterone-focused baseline within the subset where testosterone is measured.

Stage 2 (admission-level early window). An admission is labeled HF-positive if it contains ICD-10 "I50*" or ICD-9 "428*". We predict whether the admission is coded with HF using only the first 48 hours of labs from that admission. This early-window restriction reduces the risk that downstream tests or diagnoses leak information into the model and aligns with early-in-stay decision support.

C. Memory-Bounded Materialization and Ingestion Pitfalls

`labevents` is large, and full in-memory processing exceeds typical RAM constraints. We stream lab rows in chunks of 150,000 rows and apply early filtering. This enables reproducible processing under about 30 GB RAM.

In preliminary experiments, we truncated ingestion after a fixed number of rows for speed. Because the raw file is not randomized, this produces a non-random sample and can shift feature coverage and label distribution. We address this by increasing ingestion (4M vs 400M rows) and by validating downstream stability in the final pipeline. We report this explicitly because it is a common failure mode in large EHR pipelines, especially when rapid iteration leads to temporary shortcuts.

D. Feature Engineering Details

1) Testosterone channels and unit harmonization: In the temporal setting, testosterone is represented by four distinct lab channels corresponding to testosterone-related assays in `d_labitems`. These channels are kept separate to avoid conflating measurements with different clinical meaning (total vs free vs calculated vs bioavailable).

TABLE I
Testosterone-related lab channels used in the temporal tensor.

Itemid	Label	Common unit	Transform
50894	Calculated Free Testosterone	pg/mL	to ng/dL: x0.1
50988	Testosterone (Total)	ng/dL	none
50989	Testosterone, Free	pg/mL	to ng/dL: x0.1
51572	Bioavailable Testosterone	(varies / missing)	none if unclear

We also apply conservative unit harmonization. When the unit is reported as pg/mL, we convert to ng/dL using a factor of 0.1. When the unit is already ng/dL, no conversion is applied. If unit metadata is missing for an item (which can occur in subsets of extracted data), we keep the raw numeric scale for that channel rather than guessing conversions.

2) Outlier handling and imputation: Laboratory values can contain outliers due to measurement error, unit mismatch, or data entry artifacts. For Stage 1, we winsorize numeric features at the 0.5th and 99.5th percentiles and impute missing numeric values with medians. For Stage 2, we z-score values using train-only statistics and rely on explicit observation masks to represent missingness rather than aggressive imputation.

3) Static aggregations for tabular baselines: To compare sequence models with tabular baselines on Stage 2, we derive static aggregations from the temporal tensor, such as: last observed value per lab, mean over observed values, and observation counts. These features allow LR and GB to operate on the same cohort without sequence modeling, providing an interpretable reference.

4) Handling multiple measurements and temporal alignment: Within each bucket, a lab can be measured multiple times. We keep the most recent value because it is the closest to the bucket boundary and approximates the latest available patient state for that interval. We prefer the last-value strategy over mean/min/max because it reduces variance under sparsity and yields a compact representation that remains stable under limited compute.

Temporal alignment uses admission time as the reference. All labs are mapped to offsets in hours from admission. This choice reflects a common operational scenario: clinicians want risk estimates early in the admission when the care plan is still forming. It also reduces ambiguity compared to anchoring on ICU stay start, which can occur after admission and differs across pathways.

5) Mask encoding as a signal, not only missingness bookkeeping: The observation mask is not merely a technical tool to enable neural models to handle missing values. In EHR, missingness itself can be informative because it reflects clinician suspicion and protocol. Therefore, we treat masks as first-class features and evaluate explicitly whether measurement presence (for example testosterone ordered vs not ordered) carries predictive value. This design is central to our scientific conclusion: for testosterone, the measurement indicator can

TABLE II
TEMPORAL ENCODING DESIGN CHOICES FOR THE 48H LABORATORY TENSOR.

Choice	Setting
Prediction window	first 48 hours from admission
Time discretization	12 buckets of 4 hours
Within-bucket summary	most recent value (last)
Missingness representation	binary mask concatenated to values
Normalization	z-score using train-only statistics

matter more than the numeric value in a sparse setting.

E. Data Preprocessing Details

This subsection expands practical preprocessing steps that are critical for reproducibility and for avoiding subtle leakage.

1) Filtering strategy for blood-like analytes: We restrict the temporal cohort to laboratory measurements that are routinely available and are plausibly related to cardiovascular physiology. We select lab itemids with blood-like specimens (for example those typically reported in complete blood count and basic metabolic panels) and exclude non-blood fluids and point-of-care artifacts when they do not map cleanly to stable numeric ranges. This design choice prioritizes coverage and consistency across admissions, which is essential for temporal modeling.

2) Deduplication and within-bucket aggregation: Clinical labs can be repeated multiple times within a short interval. Within each 4-hour bucket and for each lab channel, we keep the most recent value, which approximates the latest available patient state for that interval. We prefer the last-value strategy over mean/min/max because it reduces variance under sparsity and yields a compact representation.

3) Train-only statistics and leakage control: All normalization statistics (mean and standard deviation for z-scoring) are computed using training data only and then applied unchanged to validation and test. This prevents information from the evaluation splits from influencing feature scaling. Similarly, the selection of the final operating threshold is performed on validation only, and then applied once on test.

4) Why missingness is modeled explicitly: Instead of aggressive imputation, we treat missingness as part of the signal and concatenate a binary mask to the value tensor. This is important because lab ordering is not random: clinicians order tests based on suspicion, protocols, and resource constraints. Explicit masks allow the model to learn whether the presence of a test (and its timing) carries predictive information beyond the numeric value itself.

F. Models

1) Logistic regression: LR provides a transparent baseline. Given feature vector $\mathbf{z}$, LR predicts $p(y = 1|\mathbf{z}) = \sigma(\mathbf{w}^\top \mathbf{z} + b)$. We use L2 regularization.

2) XGBoost and gradient boosting: XGBoost provides a strong non-linear baseline with GPU acceleration [2]. For Stage 1, it captures interactions between age, sex, and testosterone. For Stage 2, boosting is applied to static aggregations derived from the temporal tensor.

3) LSTM sequence model: We use an LSTM [5] over 12 time steps. Each step input is the concatenation of values and masks (and optionally delta-time). The final hidden state is mapped through a small MLP head. Optimization uses Adam [6].

TABLE III
EVALUATION METRICS USED IN THIS STUDY.

Metric	Why it is used
AUROC	ranking quality; can hide imbalance effects
AUPRC	focuses on positive class under imbalance [7]
F1	operating-point metric; depends on threshold
Brier	probability quality / calibration [9]

G. Split Strategy and Leakage Prevention

We split by patient identifier rather than by admission to prevent the same individual from appearing in both training and evaluation sets. This matters because repeated admissions share baseline characteristics, chronic comorbidities, and testing patterns; mixing them across splits can inflate performance. We also enforce a strict feature window (first 48 hours) so that late labs and downstream interventions do not leak post-outcome information. Finally, we compute normalization statistics on the training split only and apply them to validation and test.

H. Metrics and Operating Point Selection

Table III summarizes metrics used in this study and why they matter for imbalanced EHR outcomes.

HF is imbalanced in both stages. A fixed 0.5 decision threshold is often suboptimal. We tune the decision threshold on validation to maximize F1 and report Best-F1 with the chosen threshold. For clinical deployment, thresholds are often chosen to meet a target sensitivity (screening) or a minimum PPV (resource allocation). Accordingly, in addition to Best-F1 we report operating-point rates derived from the test confusion matrix (Sec. VI), and we include 95% confidence intervals for these rates using Wilson intervals for proportions.

I. Interpretability and Robustness

We use complementary methods:

- **SHAP for the Stage 1 XGBoost baseline** to provide quantitative feature attributions [3].
- **Ablations** that remove testosterone values, keep only testosterone flags, or add delta-time in the temporal model.
- **Group occlusion** for the temporal model by masking lab groups and measuring performance degradation.

These analyses are designed to prevent over-interpretation of sparse biomarkers and to support clinically grounded discussion.

TABLE IV
STAGE 2 TENSOR DIMENSIONS AND COMPUTE ENVIRONMENT.

Item	Value
Time steps (T)	12 (4h buckets)
Features (F)	89 labs (4 testosterone channels)
Train tensor	(381,403, 12, 89)
Val tensor	(81,819, 12, 89)
Test tensor	(82,806, 12, 89)
GPU	Tesla P100 (16GB)
RAM budget	about 30GB

TABLE V
COHORT SUMMARY (COMPACT).

Cohort	Unit	Total	HF rate
Stage 1 testosterone-measured	patients	5,673	8.57%
Stage 2 temporal labs (48h)	admissions	546,028	14.76%

IV. EXPERIMENTS

This section reports datasets, baselines, and results for Stage 1 and summarizes the temporal cohort setup that enables Stage 2.

A. Dataset: MIMIC-IV v3.1

We use MIMIC-IV v3.1 [1]. All processing is performed on de-identified data. We use patient-level splits to prevent leakage across admissions for the same individual.

B. Stage 1 Cohort: Testosterone-Measured Patients

The testosterone-focused cohort contains 5,673 patients with at least one testosterone measurement. Label distribution is HF-negative 5,187 and HF-positive 486 (8.57%). We use a 70/15/15 split: train 3,971; validation 851; test 851. Split label counts are: train (0: 3,631; 1: 340), validation (0: 778; 1: 73), test (0: 778; 1: 73).

C. Stage 2 Cohort: Early-Window Temporal Labs

The temporal cohort contains 546,028 admissions with early-window labs. We split by patient: train 381,403; validation 81,819; test 82,806. Overall HF rate is 0.1476 (14.76%). From the final test confusion matrix (Sec. VI), the test HF rate is 12,566/82,806 = 15.17%.

D. Tensor Shapes and Implementation Details

For Stage 2, the materialized tensors have shapes: train (381,403, 12, 89), validation (81,819, 12, 89), test (82,806, 12, 89), with an additional mask tensor of the same shape. Four testosterone-related channels correspond to the itemids listed in Table I. We use a GPU-enabled environment for accelerated training (Tesla P100, 16GB VRAM) and operate under a constrained RAM budget (about 30GB) enabled by streaming materialization.

E. Baselines and Hyperparameters

We use LR and XGBoost for Stage 1, and LR, GB, and LSTM for Stage 2. Representative hyperparameters are summarized in Table VI. We prioritize reproducibility and stable training rather than exhaustive search.

TABLE VI
REPRESENTATIVE HYPERPARAMETERS USED IN EXPERIMENTS.

Model	Key hyperparameters
LR (Stage 1)	L2 regularization, validation thresholding
XGBoost (Stage 1)	depth=4, estimators=300, lr=0.05, subsample=0.9
GB (Stage 2)	histogram boosting on static aggregations
LSTM (Stage 2)	12 steps, values+mask input, Adam optimizer

TABLE VII
STAGE 1 TEST PERFORMANCE WITH VALIDATION-TUNED THRESHOLDS.

Model	Thr	Acc	AUROC	AUPRC	F1
Logistic regression	0.65	0.8273	0.7396	0.2261	0.2687
XGBoost (GPU)	0.65	0.8155	0.6887	0.1641	0.2488

F. Training and Model Selection Protocol

For each model family, we follow a consistent protocol:

- Fit model parameters on the training split only.
- Select hyperparameters and the decision threshold using the validation split.
- Report final metrics on the held-out test split once.

For the deep model, we monitor validation AUPRC and apply early stopping to prevent overfitting. We then compute the F1-optimal threshold on validation and apply it to test to produce an operating-point evaluation.

G. Stage 1 Baseline Results and Thresholding Sensitivity

Stage 1 is intentionally minimal (testosterone, age, sex) and serves as a sanity check and a reference point. We note that in imbalanced settings, evaluating at a fixed 0.5 threshold can yield near-zero recall and therefore F1 near zero despite non-trivial AUROC. After tuning thresholds on validation, we obtain stable metrics in Table VII. This illustrates why operating point selection is critical in clinical ML.

H. Stage 1 Interpretability (Quantitative)

Table VIII provides gain-based importance for the Stage 1 XGBoost baseline. With only three features, these attributions are necessarily limited, but they provide a concrete view of baseline behavior.

V. STAGE 2 TEMPORAL MODELING

This section documents the temporal modeling component: early-window lab tensorization, sequence variants, and robustness analyses.

A. From Sparse Biomarker to Temporal Physiology

A limitation of the minimal baseline is reliance on a sparse biomarker. To move from a narrow hypothesis to a practical early-in-stay identifier, we broaden the input to routine labs that are more frequently measured and physiologically relevant. We model the first 48 hours of labs because this window is clinically meaningful for early risk stratification and because it limits leakage from later interventions. By including testosterone itemids in the lab panel, we evaluate its value in context rather than in isolation.

TABLE VIII
STAGE 1 XGBOOST FEATURE IMPORTANCE (GAIN).

Feature	Gain	Split count
anchor_age	1.2287	54
testo_ngdl	0.5103	141
gender	0.0109	1

B. Tensor Pipeline and Reproducibility

We implement a streaming extraction pipeline to build tensors for train, validation, and test splits under RAM constraints. We store both value and mask tensors and persist configuration artifacts (itemid lists, train statistics, and run configs) to support reproducibility and debugging. This engineering contribution enables controlled ablations and helps ensure that performance differences reflect modeling choices rather than accidental data-processing drift.

C. Sequence Model Variants

We train an LSTM on the 12-step sequence and evaluate four variants: *full* (values+mask), *no_testo* (remove testosterone numeric values), *testo_flag_only* (keep only testosterone presence flags), and *full_dt* (add delta-time). These variants isolate whether numeric testosterone values matter beyond measurement presence and other labs. To further quantify care-pathway effects, a natural extension is a global ordering baseline (mask-only vs values-only vs full) across all labs; we discuss this as a recommended analysis in Limitations.

D. Calibration and Occlusion

We measure calibration using Brier score and evaluate interpretability using group occlusion. Occlusion is computed by masking groups of related labs and measuring performance change. This provides model-agnostic evidence of which lab groups are most predictive and whether testosterone is influential in practice.

E. Reproducibility Checklist

To make results repeatable, we record: dataset version (MIMIC-IV v3.1), cohort definitions (ICD prefixes), itemid lists, time window parameters (48 hours, 4h buckets), train-only normalization statistics, and the final selected decision thresholds. We also persist model checkpoints for the selected baselines and the best LSTM variant.

VI. RESULTS AND DISCUSSION

A. Main Results on the Temporal Cohort

Table IX reports test performance for LR and GB baselines and the selected deep model. GB is strong, and the LSTM slightly improves AUPRC and Best-F1, indicating value from temporal modeling even with coarse bucketing.

TABLE IX
STAGE 2 TEST PERFORMANCE (THRESHOLD TUNED FOR BEST-F1).

Model	AUROC	AUPRC	Best-F1	Thr
LR (static agg)	0.8426	0.5563	0.5528	0.6081
GB (static agg)	0.8638	0.6295	0.5801	0.2937
LSTM (best variant)	0.8644	0.6347	0.5844	0.6460

TABLE X
STAGE 2 LSTM ABLATIONS (BEST VALIDATION EPOCH PER VARIANT).

Variant	AUROC	AUPRC	Best-F1	Thr
full (values + mask)	0.8628	0.6213	0.5789	0.722
no_testo (remove values)	0.8643	0.6222	0.5770	0.671
testo_flag_only	0.8657	0.6287	0.5816	0.656
full_dt (add delta-t)	0.8523	0.5804	0.5565	0.709

B. Ablations: Is Testosterone Helpful?

We evaluate ablations using the best validation epoch per variant (Table X). Removing testosterone values does not reduce performance. The best variant uses only a testosterone measurement flag rather than numeric values. This indicates that in MIMIC-IV, testosterone values are too sparse for population-level utility; measurement presence can reflect selection mechanisms.

C. Operating Point and Confusion Matrix

Table XI reports the test confusion matrix at the validation-selected threshold for the deep model.

To improve clinical interpretability, Table XII reports operating-point rates with 95% confidence intervals (Wilson intervals for proportions). These intervals quantify uncertainty at the chosen operating point without requiring repeated training runs.

D. Calibration and Interpretability

The deep model achieves test Brier score 0.1292, indicating reasonable probability quality [9]. Group occlusion yields physiologically plausible drivers: removing hemoglobin/hematocrit groups yields the largest AUPRC drop (about 0.09), while occluding testosterone yields near-zero impact (about 2.4 x 10^{-5}). This aligns with the ablation result and supports a cautious conclusion: testosterone numeric values do not meaningfully change predictions at population scale in this dataset setting.

E. Impact of Thresholding and Non-Random Ingestion

Two practical issues commonly degrade early EHR baselines. First, the default 0.5 threshold is often inappropriate for imbalanced outcomes and can yield near-zero recall even when ranking metrics are reasonable. Second, temporary ingestion shortcuts (for example stopping after a fixed number of rows) can bias cohort composition when the raw file is processed in fixed order. Our final protocol addresses both by enforcing patient-level splits, completing materialization, selecting thresholds on validation, and reporting operating-point performance explicitly.

TABLE XI
DEEP MODEL TEST CONFUSION MATRIX AT VALIDATION-SELECTED THRESHOLD.

	Pred 0	**Pred 1**
True 0	64,467	5,773
True 1	5,029	7,537

TABLE XII
OPERATING-POINT RATES WITH 95% CONFIDENCE INTERVALS (WILSON).

Metric	**Estimate**	**95% CI**
Sensitivity (recall)	0.5998	[0.5912, 0.6083]
Specificity	0.9178	[0.9158, 0.9198]
PPV (precision)	0.5663	[0.5578, 0.5747]
NPV	0.9276	[0.9257, 0.9295]

F. Clinical Plausibility and Error Modes

Occlusion highlights hemoglobin and hematocrit as dominant contributors, which is clinically plausible because anemia and hemodilution patterns are common in HF and correlate with renal function and volume status. Renal markers and electrolytes can reflect congestion, diuretic use, and systemic illness. These signals are broadly measured and thus contribute consistent information across the cohort.

In contrast, testosterone is rarely measured within the first 48 hours for most admissions. When present, it often reflects specific clinical contexts (for example endocrine evaluations) rather than routine HF pathways. As a result, the model can rarely leverage testosterone numeric values as a stable population-level signal. Our ablations mitigate over-interpretation by explicitly separating numeric values from ordering patterns.

G. Choosing an Operating Point for Clinical Use

Best-F1 is a useful summary in imbalanced settings, but clinical deployment typically chooses thresholds to meet target sensitivity (screening) or minimum PPV (resource allocation). Table XII provides sensitivity, specificity, PPV, and NPV at the selected threshold with confidence intervals. A practical next step is to compute thresholds that achieve a target sensitivity (for example 0.8) on validation and to report the resulting PPV and workload implications on test.

H. Generalization and External Validation

MIMIC-IV captures a specific hospital system and mix of ICU and hospital encounters. Practice patterns, lab ordering frequencies, and coding habits differ across institutions and over time. Therefore, performance measured on MIMIC-IV is not a guarantee of generalization. External validation should be prioritized, especially for biomarker-centric conclusions.

I. Baselines on Stage 2: What Each Model Captures

The Stage 2 setting highlights complementary strengths of tabular and temporal models. LR trained on static aggregations is an interpretable reference that captures coarse associations between early labs and HF labels, but it cannot model non-linear interactions or temporal dynamics. Gradient boosting captures non-linear effects and interactions, explaining strong baseline performance. However, aggregation-based models discard ordering information within the 48-hour window.

The LSTM consumes the discretized sequence directly (values and masks over 12 steps). Even with coarse 4-hour buckets, sequence input can encode whether abnormal labs appear early vs late, whether values recover, and whether testing density changes over time. These patterns reflect both physiology and care processes. The modest gains over boosting suggest that much of the signal is present in static summaries, and that temporal improvements are limited by sparsity and coarse discretization.

J. Error Analysis (Qualitative)

False positives can arise when comorbidities produce laboratory patterns that resemble HF physiology, such as renal dysfunction, anemia, or systemic illness that triggers dense testing and abnormal electrolytes. In these cases, the model can identify elevated risk but may assign it to an HF-coded admission label that is not present.

False negatives can occur when early labs do not capture HF-related changes, either because testing is sparse in the first 48 hours or because HF is coded based on history rather than acute decompensation in the current stay. Another pattern is delayed physiologic deterioration after the early window; our strict 48-hour restriction avoids using later information, so these cases are expected to be harder.

K. Ethical and Deployment Considerations

This study is positioned as decision support rather than automated diagnosis. In deployment, predictions should be used to prompt additional review, prioritize follow-up, or trigger checklists, not to replace clinician judgment. Because EHR models can learn care-pathway effects, hospitals should validate in local practice, monitor drift over time, and recalibrate probabilities when protocols or patient mix change.

Fairness considerations are also relevant. Sex and age correlate with both HF prevalence and lab ordering. In a testosterone-centered study, measurement is selective, and naive use could amplify disparities if predictions depend on who gets tested. Our design partially mitigates this risk by explicitly modeling missingness and by reporting ablations showing limited marginal effect of testosterone numeric values at population level.

L. Effect of Delta-Time Features under Sparse Sampling

In principle, time-gap features can improve missingness-aware models by distinguishing recently observed values from stale values. In our setup, adding delta-time (variant full_dt) reduces validation performance. A plausible explanation is that delta-time estimates are noisy under heavy sparsity and coarse bucketing, and the model can overfit to irregularities in ordering rather than physiology. This suggests that more careful time-gap modeling may be needed if delta-time features are used [4].

TABLE XIII
KEY LIMITATIONS AND MITIGATION STRATEGIES.

Limitation	Mitigation / recommendation
ICD label noise and timing ambiguity	Early-window restriction; consider adjudicated labels in future
Sparse testosterone measurements	Evaluate flags vs values; analyze testosterone-measured subcohort
Missing not at random	Include masks; interpret cautiously; consider missingness-aware models
Threshold sensitivity under imbalance	Tune on validation; report operating-point metrics and CIs
Single-institution data	External validation; robustness to practice-pattern shifts

M. Selection Effects as a First-Class Modeling Concern

The testosterone case study highlights an important lesson for EHR ML: ordering behavior is often predictive. If a test is ordered only when clinicians suspect endocrine or reproductive issues, the measurement indicator can correlate with comorbidity profiles and care pathways. Our ablations separate these effects by comparing numeric values vs flags. The near-zero occlusion impact of testosterone values and the competitiveness of flag-only variants indicate that, in this dataset, testosterone primarily acts as a proxy for selection rather than a direct physiological driver.

N. Threats to Validity

Several factors limit the generality of our findings. **Task definition:** HF labels are ICD-derived and can reflect pre-existing disease; onset time is not modeled. **Label noise:** ICD codes can misclassify HF and do not precisely encode onset time. **Selection bias:** measurement patterns reflect clinical workflows; testosterone is especially selective. **Domain shift:** models trained on MIMIC-IV may not transfer to other hospitals. **Unobserved confounding:** comorbidities and treatments may drive both labs and HF codes, complicating interpretation.

To quantify care-pathway effects more broadly, an informative additional analysis is a global ordering study: (i) mask-only model (presence/absence of labs), (ii) values-only model (imputed values without masks), and (iii) full model (values+mask). This would measure directly how much signal can be obtained from ordering patterns alone.

O. Mitigation Strategies and Practical Recommendations

Based on lessons learned, we summarize practical recommendations for EHR modeling: **(i) enforce group splits** when repeated measurements per patient exist, **(ii) restrict features to a clinically justified prediction window**, **(iii) report imbalance-aware metrics and tune thresholds on validation**, **(iv) include model-agnostic interpretability tests** (ablations or occlusion), and **(v) document any data-processing shortcuts** used during early debugging.

P. Limitations and Practical Implications

(1) We identify HF-coded admissions using early labs; HF onset time is not modeled and HF may represent pre-existing disease. (2) Testosterone is sparse and ordered in selective contexts, limiting population-level incremental value. (3) We do not claim causal effects of testosterone or other biomarkers. (4) External validation is needed to assess generalizability across institutions.

VII. CONCLUSION

We developed an end-to-end HF identification pipeline in MIMIC-IV, progressing from a testosterone-focused baseline to early-window temporal lab modeling. Baselines show non-trivial discrimination but are sensitive to imbalance and decision thresholding, motivating explicit operating-point reporting. Our temporal lab representation and LSTM achieve strong performance (test AUROC 0.8644, AUPRC 0.6347) and enable ablation-driven conclusions: testosterone numeric values provide near-zero marginal gain at population level in this EHR setting, while measurement patterns can encode selection effects. Future work should evaluate a dedicated testosterone-measured subcohort, quantify global ordering effects (mask-only vs values-only vs full), incorporate additional modalities, and validate externally across institutions with different testing practices.

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